Data Structures and Complexity

Big-O notation describes algorithm time complexity. Array access O(1), search O(n). Binary search O(log n). Merge sort O(n log n). Bubble sort O(n^2). Linked lists allow O(1) insertion at head but O(n) search. Hash tables provide O(1) average lookup. Binary Search Trees offer O(log n) operations when balanced. Graphs use BFS and DFS for traversal.